26 $KE = \frac{1}{2}mv^2 = \frac{p^2}{2m}$

Loss in EPE = Gain in KE

$$eV = \frac{p^2}{2m} \Rightarrow p = \sqrt{2meV}$$

$$\lambda_{deBroglie} = \frac{h}{p} = \frac{h}{\sqrt{2meV}}$$

$$c = f_{photon}\lambda_{photon} \Rightarrow \lambda_{photon} = \frac{c}{f_{photon}}$$

For photon, $\lambda_{photon} = \lambda_{deBroglie}$

$$\frac{c}{f_{photon}} = \frac{h}{\sqrt{2meV}} \Rightarrow f_{photon} = \frac{c\sqrt{2meV}}{h}$$

Answer: A

27 When the atoms of the cool vapour absorbs a photon, it becomes excited and is unstable.